A Dual-IMU Barbell Velocity Tracker

Measuring barbell velocity for velocity-based training is important for powerlifters. We present a system using commonly available sensors, showing an accuracy close to state-of-the-art existing systems.

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Introduction

Velocity-based training (VBT) involves calculating the linear velocity during strength training exercises. The utilization of VBT has been shown to produce gains in maximal strength in the back squat, bench press and deadlift that are at least as large as traditional percentage-based loading, despite lower training loads [1]. It can also be used to estimate one-repetition maximum load [2]. Coaches, however, report difficulties with using VBT, including the cost, ease of use and accuracy of the measuring devices [3].

One study evaluated three existing devices for measuring peak and mean velocity [4]. It found that the linear position transducer was the most sensitive device, but highlighted that it was expensive and that the cable attachment could interfere with the lift. The study also compared two devices utilizing inertial measurement units (IMUs) that are cheaper and wireless but found them to be less accurate. Part of the challenge with IMUs is that velocity must be obtained by integrating acceleration, which accumulates drift over time. The barbell also tends to tilt and rotate during a lift, so the accelerometer axes do not stay aligned with the direction of movement, and acceleration alone cannot separate this rotation from linear motion.

The contributions of this paper are an approach to VBT using two commonly available *Movesense* IMU sensors, mounted on the ends of the barbell and a web platform for communicating the results to athletes and coaches to guide training. Our research question is: can two end-mounted IMUs estimate the mean and peak velocity of each repetition with a mean absolute error below 0.10 m/s, small enough that a repetition is not placed in the wrong training zone?

Design

In this paper a complete system for VBT is built and evaluated. The system architecture is described in Figure 1. The system uses the commercially available *Movesense* sensor. The sensor is equipped with a 9-axis IMU and radio communication through Bluetooth Low Energy [5]. It was chosen over a custom IMU board, which would have required us to build the firmware, Bluetooth stack and enclosure ourselves, reducing the time available for analysis.

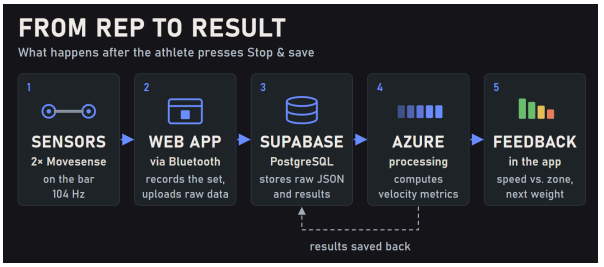


Figure 1: System Architecture

The two sensors stream data at 104 Hz over Bluetooth to the receiving device. The raw data is stored as a JSON in Supabase, an open-source backend platform built on PostgreSQL, together with the set's repetitions, load and target velocity zone. Each new recording triggers a serverless processing step in Microsoft Azure through a database webhook. It reads the raw data, computes the velocity metrics, and stores the results in the same database, from which the application reads and displays them along with feedback for the next set.

To estimate the velocity the acceleration and gyroscope data is filtered using the Madgwick filter [6]. After filtering the vertical acceleration data is used to estimate the velocity. The velocity drifts over time, to account for this a linear approximation of the drift is calculated. The drift is used to correct the velocity as illustrated in Figure 2. The peaks in the velocity are used to characterize each repetition, all velocity data above 0 around each peak is used to estimate the mean velocity. When extracting the peaks the peak height decides the peak velocity.

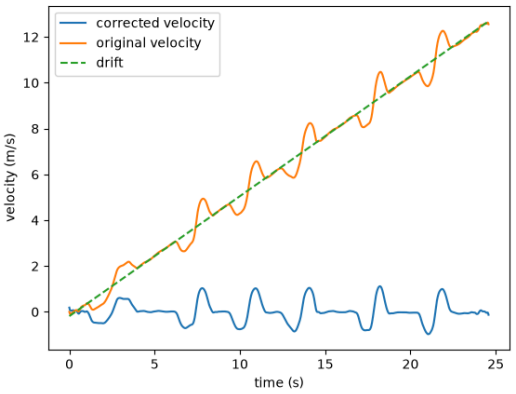


Figure 2: Example of velocity drift for six squats

The web platform is built to have two different views, one for coaches and one for athletes. A team can consist of multiple coaches and athletes where the coaches assign workouts for the athletes and can see the results from each athlete's workouts.

When starting the workout the athlete or the coach adds the exercise they are going to do, the number of repetitions, intended speed zone and the weight of the barbell into the athlete's view. When ready to perform the lift the athlete lifts the barbell up and the coach starts the recording. After the lifts are performed the coach stops the recording before the athlete puts the barbell back.

After each recorded set, the athlete sees the mean and peak velocity of every repetition. If the set was performed outside the intended speed zone, the application suggests increasing or decreasing the load for the next set, based on the athlete's own load-velocity profile; if it was inside the zone, it suggests keeping the load. All workouts are stored in the platform's database and are available to both the athlete and their coach.

Evaluation

To validate the velocity estimations 4 participants (2 females aged 24 and 26 and two males aged 23 and 24) performed 6 squats in 3 different speed zones defined by the peak velocity, v, of the barbell. The participant group was not powerlifters but former athletes who do strength workouts 1-4 times per week. The speed zones were divided into starting strength ($v > 1.3$ m/s), speed strength ($0.5 < v < 1.3$ m/s) and absolute strength ($v < 0.5$ m/s). When performing the squats the velocity was recorded simultaneously using the *Movesense* sensor and the Gymaware RS device ("the golden standard") [7]. The estimated velocity from the *Movesense* data and the Gymaware result was compared by calculating the mean absolute error and standard deviation when comparing the results for both mean velocity and peak velocity. The most accurate result was obtained for the absolute strength with a mean absolute error of $0,03 \pm 0,02$ for mean velocity and $0,04 \pm 0,02$ for peak velocity. The least accurate results were for the starting strength with a mean absolute error of $0,33 \pm 0,17$ for mean velocity and $0,21 \pm 0,13$ for peak velocity. For speed strength the standard error for mean velocity was $0,05 \pm 0,16$ and $0,06 \pm 0,03$ for peak velocity.

The large error in the starting-strength zone is of limited practical consequence for the intended users, since powerlifting training is predominantly performed at heavier loads and lower velocities. It does, however, show that the method does not generalize to explosive, light-load training without further work.

Figures 3 and 4 illustrate the difference between the Gymaware results and the estimated velocity using *Movesense* for absolute strength. One limiting factor in the accuracy of velocity calculations is the linear drift approximation. Theoretically, to obtain better results a more accurate approximation of the drift could be used. The linear approximation works best if the drift is continuous in one direction, when the drift changes direction a quadratic approximation could give a better result.

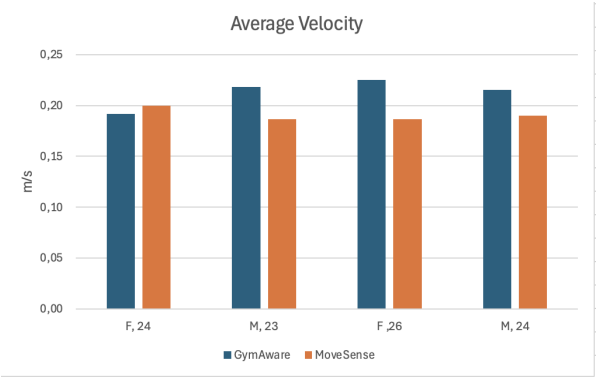


Figure 3: Average velocities for absolute strength for all 4 participants. The average value across all 6 repetitions averaged for each participant, both for the Gymaware values and the Movesense calculated values.

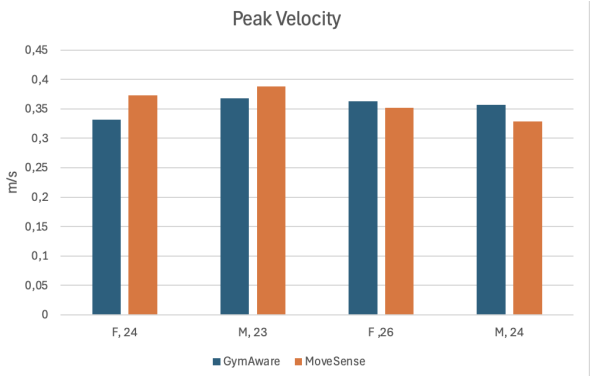


Figure 4: Peak velocities for absolute strength for all 4 participants. The average value across all 6 repetitions averaged for each participant, both for the gymaware values and the calculated values.

Besides the velocity accuracy, the feedback loop of the platform was tested. In a preliminary test of seven recorded sets, processing took 0.9–7.5 s (median 5.8 s), measured from the recording being stored in the database until its results were available. The total time from stopping the recording to the feedback appearing additionally includes the upload of the raw data.

There are some clear limitations in the evaluation process. The testing was focused on the squat exercise and not deadlift and bench press. There is a possibility that these exercises have a reduced accuracy. Another possible limitation is that the participants of the testing were not powerlifters, they were all accustomed to strength workouts but not at a professional level. The level of quality in the technique during the performed squats were not evaluated and can therefore differ from when professional powerlifters perform the squats and therefore affect the movements of the barbell.

Conclusion

Compared with gold-standard equipment, the *Movesense*-based estimates met the 0.10 m/s criterion for squats at low and moderate velocities, but not at high velocities, where the error in mean velocity grew to 0.33 m/s. Peak velocities above 1.3 m/s only occur with very light loads, which are rarely used in powerlifting; for the loads that dominate powerlifting training, the system is accurate enough to guide training.

We presented a lower-cost barbell velocity tracker that uses two *Movesense* sensors mounted on the ends of the barbell and combines accelerometer and gyroscope data to estimate velocities together with a web platform for showing the results to athletes and coaches. Limitations are the small number of test participants and the noise sensitivity. Future work includes improved drift handling to extend the accuracy to fast, light lifts, the bench press and deadlift, a larger validation study with powerlifters, more ways to use the data, and support for other IMU sensors.

Github project repository

https://github.com/SportsTechVeloProject/.github

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